

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

B. ENG. DEGREE EXAMINATIONS. SECOND SEMESTER 2024/2025 SESSION

COURSE CODE:EEE 512/ICT 516 **COURSE TITLE:** Radio Frequency Spectrum

Management

UNITS: 3

TIME ALLOWED: 3 hrs

Instruction: Answer any four (4) questions. Carefully read all questions, provide equations and diagrams where needed.

Question 1 (15 marks)

- (a) Define spectrum sharing and state why it is important in modern wireless communication (2 marks)
- (b) Use simple sketch to show the basic elements of spectrum management (5 marks)
- (c) Itemize four roles of spectrum management in the ICT sector. (2 marks)
- (d) List six functions each of a spectrum planning and spectrum engineering personnel (6 marks)

Question 2 (15 marks)

- (a) State the aim of spectrum sensing, spectrum sharing and spectrum mobility. (3 marks)
- (b) State the reason why spectrum scarcity arises despite large licensed allocations (2 marks)
- (c) Discuss three major challenges with spectrum sensing. (6 marks)
- (d) List 2 national and 6 international spectrum regulatory body. (4 marks)

Question 3 (15 marks)

- (a) What do you understand by the following terms;
 - (i) Spectrum trading (1.5 marks)
 - (ii) Network slicing (1.5 marks)
 - (iii) Spectrum repurposing and (1.5 marks)
 - (iv) Spectrum refarming. (1.5 marks)
- (b) Explain the various trading mechanism design for spectrum management (6 marks)
- (c) Outline two ways spectrum sharing promotes innovation? (3 marks)

Question 4 (15 marks)

- (a) What is white space device (2 marks)
- (b) A TV station operating between 487.25MHz to 493.25MHz in a small urban area has a transmitter with a rated power of 40 kW with effective radiated power of 24 kW. The transmitting tower height is 300 m with transmitter gain of 31.70 dBi. At 4 km away, a spectrum analyzer with a gain of 2.25 dBi is located for energy detection at a height of 10 m. Taking fading margin to be 1 dB, SINR to be 15 dB, and an optimistic pathloss of 120, determine the following parameters needed to guarantee primary users are protected from interference from secondary devices;
 - (i) Pathloss (5marks)
 - (ii) Protection radius (3 marks)Given that $N_o = 10\log(KTB)$.
- (c) State the importance of spectrum sensing in cognitive radio networks (2 marks)
- (d) List three applications of TVWS. (3 marks)

Question 5 (15 marks)

- (a) What is television white space (TVWS) (3 marks)
- (b) List and briefly discuss two major challenges faced in implementing spectrum sensing techniques (4 marks)
- (c) Compare energy detection and cyclostationary feature detection as spectrum sensing techniques. (6 marks)
- (d) State four advantages of TVWS over Wi-Fi technology. (2 marks)

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Question 6 (15 marks)

- (a) Explain why spectrum management is an important tool for governments to optimize the use of finite public resources. (4 marks)
- (b) State how spectrum availability drives smart infrastructure development. (2 marks)
- (c) State how cooperative spectrum sensing helps to improve detection reliability. (4 marks)
- (d) Explain the reasons why spectrum underutilization and scarcity has become the order of the day. (5 marks)

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